

BEALE AFB, CA, USA

WMO: 724837

Lat: 39.133N

Lon: 121.433W

Elev: 113

StdP: 14.64

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 93216

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	31.1	33.8	20.6	15.5	48.1	24.2	18.4	47.2	29.9	53.0	25.9	53.8	3.6	90	0.457

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.5	101.2	70.1	98.4	69.3	95.4	68.1	73.1	95.7	71.3	93.3	69.9	91.4	7.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.0	92.9	82.8	63.3	87.3	80.0	61.4	81.8	78.7	36.6	95.6	35.1	93.8	33.8	91.4	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.5	19.5	17.0	DB	25.9	105.8	3.6	1.5	23.4	106.9	21.3	107.7	19.3	108.6	16.7	109.7
			WB	25.3	76.2	3.6	2.1	22.7	77.7	20.6	78.9	18.6	80.1	15.9	81.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.6	48.0	51.4	55.3	59.9	66.5	74.0	78.6	77.3	73.7	64.0	54.2	47.0	(6)
	DBStd	12.30	4.84	4.85	5.10	6.05	6.49	6.61	4.67	4.69	5.44	5.05	5.56	5.36	(7)
	HDD50	294	94	38	12	3	0	0	0	0	0	0	22	125	(8)
	HDD65	2413	526	380	302	174	57	7	0	0	5	78	325	559	(9)
	CDD50	4877	33	78	176	301	511	720	887	847	712	435	147	30	(10)
	CDD65	1523	0	0	2	21	103	278	422	382	266	48	1	0	(11)
	CDH74	20536	0	1	56	434	1481	3789	5646	4972	3331	795	31	0	(12)
	CDH80	10742	0	0	5	115	610	2046	3259	2772	1691	242	2	0	(13)
(14) Wind	WSAvg	6.8	6.5	7.5	7.7	7.6	7.6	7.3	6.7	6.2	6.1	6.0	5.9	6.7	(14)
(15) Precipitation	PrecAvg	26.9	5.3	4.5	4.2	1.8	1.3	0.4	0.0	0.1	0.2	1.3	2.3	5.4	(15)
	PrecMax	44.0	14.2	12.2	13.4	6.2	4.7	1.6	0.2	0.9	0.7	4.1	5.6	12.2	(16)
	PrecMin	8.9	0.5	0.0	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.6	(17)
	PrecStd	8.0	3.6	3.2	3.7	1.5	1.3	0.5	0.0	0.2	0.2	1.1	1.6	3.5	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.5	71.9	80.3	88.1	96.8	104.3	104.3	104.1	100.6	91.5	77.5	64.2	(19)
		MCWB	52.8	57.8	61.6	63.8	69.4	71.8	71.3	70.0	70.0	63.8	59.3	53.8	(20)
	2%	DB	62.1	68.0	74.8	83.4	91.0	99.8	100.8	99.9	96.5	86.1	73.4	61.0	(21)
		MCWB	53.0	56.7	59.2	63.0	65.5	70.0	70.0	69.6	67.2	62.4	58.0	53.3	(22)
	5%	DB	59.3	64.1	71.3	79.4	87.1	95.5	98.0	96.5	93.3	82.0	69.3	58.6	(23)
		MCWB	53.1	54.7	57.6	61.1	64.6	67.8	69.4	68.3	66.3	60.4	56.2	52.6	(24)
	10%	DB	56.9	61.1	67.5	74.8	82.2	91.2	94.8	93.3	90.1	78.6	64.6	56.1	(25)
		MCWB	51.9	53.7	55.9	59.4	62.8	66.4	68.9	67.5	65.3	59.2	55.0	51.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.2	61.1	63.8	68.1	71.9	75.4	75.6	73.5	72.4	67.0	62.6	59.2	(27)
		MCDB	60.3	67.2	74.5	82.2	90.5	98.3	97.4	98.5	95.7	83.7	69.9	60.6	(28)
	2%	WB	56.5	58.9	61.1	64.4	68.8	72.5	73.2	71.5	69.7	64.4	60.5	56.8	(29)
		MCDB	59.3	64.2	71.2	79.4	86.8	93.8	96.0	95.2	90.7	80.3	68.2	58.6	(30)
	5%	WB	54.5	57.0	59.1	62.3	66.2	70.3	71.4	70.0	68.0	62.7	58.8	54.6	(31)
		MCDB	57.0	62.0	67.9	76.3	83.6	90.8	93.3	92.7	89.2	76.9	65.7	56.6	(32)
	10%	WB	52.9	55.1	57.2	60.4	63.9	68.2	69.8	68.5	66.3	61.1	56.9	52.7	(33)
		MCDB	55.4	59.6	65.0	72.6	79.5	88.2	91.7	90.4	86.9	74.0	63.1	54.7	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	15.6	17.9	20.6	24.3	25.9	29.4	31.5	31.2	29.6	26.6	20.1	16.1	(35)
		MCDBR	20.5	23.8	28.4	32.3	32.8	34.5	34.2	34.2	34.5	32.8	27.2	19.3	(36)
	5% WB	MCWBR	13.1	14.1	15.2	15.1	13.0	12.7	13.6	13.8	13.7	14.2	13.7	12.0	(37)
		MCWBR	13.4	19.2	24.2	28.9	30.2	32.3	32.0	32.3	31.5	27.1	20.6	11.8	(38)
(40) Clear-Sky Solar Irradiance	taub		0.323	0.326	0.350	0.358	0.371	0.363	0.361	0.345	0.353	0.345	0.324	0.335	(40)
		taud	2.505	2.505	2.429	2.401	2.391	2.421	2.415	2.454	2.440	2.462	2.510	2.454	(41)
	Ebn at Noon		270	285	288	291	288	289	289	292	283	274	267	254	(42)
		Edh at Noon	25	28	33	36	38	36	36	34	33	29	24	24	(43)
(44) All-Sky Solar Radiation	RadAvg		679	987	1407	1885	2314	2612	2616	2336	1880	1307	821	595	(44)
	RadStd		123	134	164	178	154	114	81	78	70	96	68	103	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (41 neighbors)		+0.74	N/A	-2.03	N/A	N/A	N/A	N/A	-193	+247	+79	

Nomenclature: See separate page